

Wednesday July 8th		Thursday July 9th	Friday July 10th
9:00-10:30	09:00-09:15 - Registration - 9:15-9:30 - Welcome		S07 - Neuromusculoskeletal & Sensory Control - Chairs: Bart Bijnsen, Renate Janssen
	09:30-10:30 - Opening Keynote - Blanca Jordan , Projects Lead for Strategic Innovation, AnySolution - The Uncharted Path: Professional success, failure, continuous training, personal work, and pasion, along professional growth - Blanca Jordán is a senior executive with 25 years of experience driving strategic innovation and digital transformation across the healthcare and life sciences sectors. Throughout her diverse career, she has held leadership roles as CTO, COO, and CPO, bridging the gap between business growth and deep technical environments. Recognized for her applied research background and natural leadership, she has served as a key expert for the European Commission and actively steers international technology roadmaps. Blanca is a firm believer in continuous learning, combining a background in Telecommunications Engineering with degrees in Sociology and Psychology.(https://www.linkedin.com/in/blancajordan/) 	S04 - Medical Devices - Chairs: Laura Becerra, Albert Hernansanz 09:30-09:55 - BT 09 - Irene Flores González - Design and Manufacturing Pipeline for Personalized Ostomy Systems in the Management of Enteroatmospheric Fistulas 10:00-10:25 - BT 11 - Esther Ortiz Rejón - Non-Invasive Framework for Menstrual Cycle Monitoring Using Saliva Ferning Patterns and Machine Learning 10:30-10:55 - BT 12 - Ricard Más Secanell - Prototype design, fabrication, and validation for preclinical evaluation of a non-battery active implantable medical device	09:00-09:25 - BT 14 - Alba Barnes Calle - AI-Based Computer Vision System for Post-Stroke Upper Limb Rehabilitation 09:30-09:55 - BT 15 - Izaskun Sedano Martínez - Development of a Process to Validate the Design for a Hybrid Lower-Limb Exoskeleton Integrating Functional Electrical Stimulation 10:00-10:25 - BT 16 - Laura Ovelheiro Barreiro - External Unit Prototype for a Minimally Invasive Neuromodulation System for Chronic Headaches: Towards Preclinical Large-Animal Studies 10:30-10:55 - BT 13 - Aina Bennassar Pertusa - Toward individualized medicine in chronic stroke: diffusion MRI biomarkers of CST integrity and motor network reorganization for personalized neuromodulation
	10:30-11:00 Coffee		
11:00-13:00	S01 - Methods in Biomedical Imaging - Chairs: Oscar Camara, Gonzalo Plaza 11:00-11:25 - BT 01 - Emily Ji Lam - Vascular Segmentation and Characterisation of Porto-Sinusoidal Vascular Disease using Synchrotron-based X-Ray Phase Contrast Imaging 11:30-11:55 - BT 02 - Jana Casau Pueyo - Generation and Validation of Synthetic Brain Computed Tomography Images for Ischemic Stroke 12:00-12:25 - BT 03 - Laiba Mahmood Ayesha - Automatic Segmentation of the Globus Pallidus Internus in Paediatric Dystonia Using Conventional MRI 12:30-12:55 - BT 04 - Albert Puga Sánchez - Monocular Tissue Synthesis from Endoscopy via Triangle Splatting	11:00-11:30 Coffee S05 - Cardiovascular Functions, Disorders & Characterization - Chairs: Eva Casoni, Antoni Ivorra 11:30-11:55 - BT 25 - Arnau Martínez Pérez - 3D Characterization of Cardiac Microstructure in X-ray Phase Contrast Images of Endomyocardial Biopsies using Machine Learning 12:00-12:25 - BT 26 - Maria Cortés Díaz - Towards Connectivity and Geometry Aware Loss Optimization for Intracranial Vessel Segmentation in Stroke Imaging 12:30-12:55 - BT 27 - Lola Carcereny Martínez - Modeling Flow Diverter Expansion in Endovascular Treatment of Intracranial Aneurysms 13:00-13:25 - BT 28 - Iris Azlor Solé - Diagnostic performance of EPAC1 in CAD in the context of T2DM 13:30-14:30 Lunch	11:00-11:30 Coffee S08 - Biomedical Signal Complexity & Analysis - Chairs: Kathakali Sarkar, Demetrio Ferro 11:30-11:55 - BT 37 - Madeleine Fairey - Characterising Synchronisation on Ultra-Long Timescales in Epilepsy using Phase-Based Analysis on Subcutaneous Electroencephalographic Recordings 12:00-12:25 - BT 36 - Marc Adrover Pons - Beyond Apparent Structure in Ultra-Long-Term Electroencephalography Recordings of Epilepsy Patients 12:30 - 12:55 - BT 38 - Alex Just Barbero - Nonlinear Frequency Mixing as a Mechanism for the Generation of Infra-Slow Oscillations in Coupled Neural Mass Models 13:00-14:30 Lunch
	13:00-14:30		
14:30-16:00	S02 - Biomechanics - Chairs: Elham Alizadeh, Abi Moslemivaraki, Estefano Muñoz-Moya 14:30-14:55 - BT 18 - Anna Freixas Morato - Orthosis Design for Climbers with Hand Arthrosis 15:00-15:25 - BT 19 - Ruth Agudo Belmonte - Standardized Yoga Protocol: Biomechanical Characterization of Modalities and Expert Motor Strategies 15:30-15:55 - BT 20 - Xenia Yan Hueso Espinosa - Effect of Prolonged Load on Knee Articular Cartilage's Biomechanical Response	S06 - Imaging & Modelling in Clinics - Chairs: Gabriel Bernardino, Marc Molla 14:30-14:55 - BT 22 - Blanca Dongmei Martínez Piñero - Development and Evaluation of a 3D-Printed Anatomical Model to Improve Doctor-Patient Communication in Surgical Procedures 15:00-15:25 - BT 23 - Carla Paires Ramis - Prediction of atrial fibrillation in hypertrophic cardiomyopathy using atrial morphology derived from cardiac magnetic resonance images 15:30-15:55 - BT 24 - Alba Valero Viñals - Machine learning model integrating longitudinal MRI-Radiomics, Genetic Biomarkers, and Clinical Data for Outcome Prediction in Glioblastoma Patients undergoing Stupp Protocol 16:00-16:25 - BT 21 - Anna Gallart Serracanta - Development of an Advanced Surgical Simulation Model for the Anterior Approach to the Hip	S09 - Neuroimaging & Signalling - Chairs: Demetrio Ferro, Oscar Camara, Yusufbaran Tanriverdi 14:30-14:55 - BT 32 - Elena Polls Fuertes - Deep Brain Stimulation for Chronic Cluster Headache: A Structural Connectivity Study 15:00-15:25 - BT 33 - Laura Montserrat Sokorina - Characterization of Thalamic Nuclei in Patients Receiving MRgFUS Treatment for Essential Tremor: A Radiomic-based study 15:30-15:55 - BT 34 - Alex Trasancos Luque - Normative Amyloid-Tau PET Abnormality Mapping for Alzheimer's Disease Staging and Cognitive Impairment Classification 16:00-16:25 - BT 35 - Anna Tocabados Gassol - Idiopathic Generalized Epilepsy: Interictal EEG Connectivity as a Biomarker of Disease Severity and Cognitive Dysfunction 16:30 Picture + Farewell Aperó
	16:00-16:30 Coffee		
16:30-18:00	S03 - Biological Systems Biophysics & Modelling - Chairs: Zerihun Workineh, Katherine Crump 16:30-16:55 - BT 29 - Lucia Alcalde Calvete - Influence of Scaffold Porosity Gradients on Cell Differentiation 17:00-17:25 - BT 30 - Nil Guell Melgar - Design and Implementation of a Low Intensity Focused Ultrasound Setup to Induce Mechanosensible Calcium Response in Microglial Cells 17:30-17:55 - BT 31 - Lucas Pellicer López - Engineering arabinose-inducible recombinase-controlled CRISPRi oscillators in Escherichia coli for synchronized population-level dynamics		CONFIDENTIAL PRESENTATIONS